Pranav Achar

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Robot learning, computer vision and AI agents. Seeking Summer 2027 software and machine learning internships.

Education

Los Angeles Mission College, A.S. Computer Science & Mathematics (transfer track) **Expected transfer: Fall 2027**
4.0 GPA at LA Mission College. Also coursework at Diablo Valley College and Coastline College. 2026 to present
Coursework: Calculus I and II, Programming in C++; *in progress:* Calculus III, Advanced C/C++; *Spring 2027:* Linear Algebra & Differential Equations, Discrete Mathematics, Data Structures.

Projects and Awards

Understudy [code] [live] | **1st Place**, Runway API Hackathon ($10,000 prize) Sep 2026
- Turns one sentence into Runway-generated, physics-checked robot demonstrations and trains a vision-language-action model (SmolVLA) on a rented GPU. In simulation, the policy scored 43/50 vs 44/50 for perfect scripted demos.

Player Two [code] [live] | **1st Place**, Executable World AI Hackathon; **2nd Place**, Battle of the Personal Brains Sep 2026
- Built an agent that turns a plain-language task into robot training data: it finds Creative Commons video, tracks pose with MediaPipe, and retargets it onto Unitree G1, SO-101 and Franka Panda robots via inverse kinematics in MuJoCo.
- One unattended run screened 68 videos and accepted 22 in under 17 minutes, exporting a LeRobot v3.0 dataset behind deterministic gates. Rebuilt as an AWS Strands agent with Cognee memory and a Docker sandbox.

Foundry [code] [live] | **1st Place**, Superserve Track, terac One-Person-Company Hackathon Aug 2026
- Built an agent that reads a real audience, writes a business for each person in it, then builds and deploys every storefront and runs outreach on a schedule. Spend ceilings, an append-only ledger and allowlisted outreach are enforced in code.

Finalist (top 6 of 115), Abridge x Anthropic x Lightspeed Hackathon; **1st Place**, Apify Track, Build Club Jul 2026

SixthSense [code] | **2nd Place**, ExecuTorch Hackathon (Qualcomm x Meta) Jun 2026
- On-device navigation copilot for blind users, featured on the PyTorch Foundation blog. Built the ExecuTorch vision pipeline and LLM voice agent on a Galaxy S25 that runs Depth-Anything-V2, YOLOv11, Qwen2.5 and OCR fully offline.

HealthFlow [code] | **1st Place**, Scalekit x Apify Hackathon: 9-agent voice-to-EHR pipeline. May 2026

Research

Independent Researcher, vision-language models (VLMs) for camera-based event monitoring Jul 2026 to present
- Researching vision-language models that run on the camera device, notice and flag observable events in video (a fall, a person down) and save reviewable clips. VLM work in progress.
- Pose-based prototype fall model on a public held-out benchmark it never trained on (UR Fall): flagged all 14 scored falls, a median 2.6 s after onset. 100+ GPU hours of training.

Independent Researcher, markerless motion capture measurement study Sep 2026 to present
- Designing a preregistered study of how skip-frame person detection, a speed-up option, shifts joint angles in open-source markerless motion capture (rtmlib, RTMPose). In progress; preprint (not peer-reviewed) planned for November 2026.
- Traced the cause in rtmlib's code: between detections the person box is rebuilt from the last frame's keypoints, which snap to a 0.5-pixel grid, so the loop never settles; freezing the box makes the output identical. Pilot (10 still photos): skip-frame detection shifted joint angles 4.2° vs 1.1° from invisible image noise alone.

AI Research Intern, UC Santa Cruz AIEA Lab: multi-step logical reasoning in LLMs. May 2026

Experience

Co-Founder and CEO, Web Design Initiative (nonprofit) 2024 to May 2026
- Founded and led a 6-engineer volunteer team building pro-bono websites (8+ clients in year one), including Peninsula Tamil School (bilingual English/Tamil), the cybersecurity nonprofit CPFA and a local lab-supply distributor.

Engineering Intern, Opterra Energy Solutions Jun to Aug 2026
- Designed the system and built the financial model for a Heat-Recovery-as-a-Service proposal to sell AI data center waste heat as district heat (about $3.5M/yr per 10 MW site, 2 to 4 year payback), and wrote the proposal.

Open Source

18 merged pull requests to statsmodels (14), Biome (2), LanceDB (1) and Lightning CSS (1): 11 statsmodels bug fixes (32M PyPI downloads a month; two issues open since 2018), GLM docs, a LanceDB client fix, and CSS-parser fixes.

Skills

Languages: Python, TypeScript, JavaScript, C++, Kotlin, Java, SQL, Rust
ML and robotics: PyTorch, ExecuTorch, pose estimation (MediaPipe, RTMPose), YOLO, MuJoCo, LeRobot, NumPy
Agents and systems: Anthropic SDK (Claude), LangGraph, LangChain, Next.js, React, Node.js, Postgres, Docker